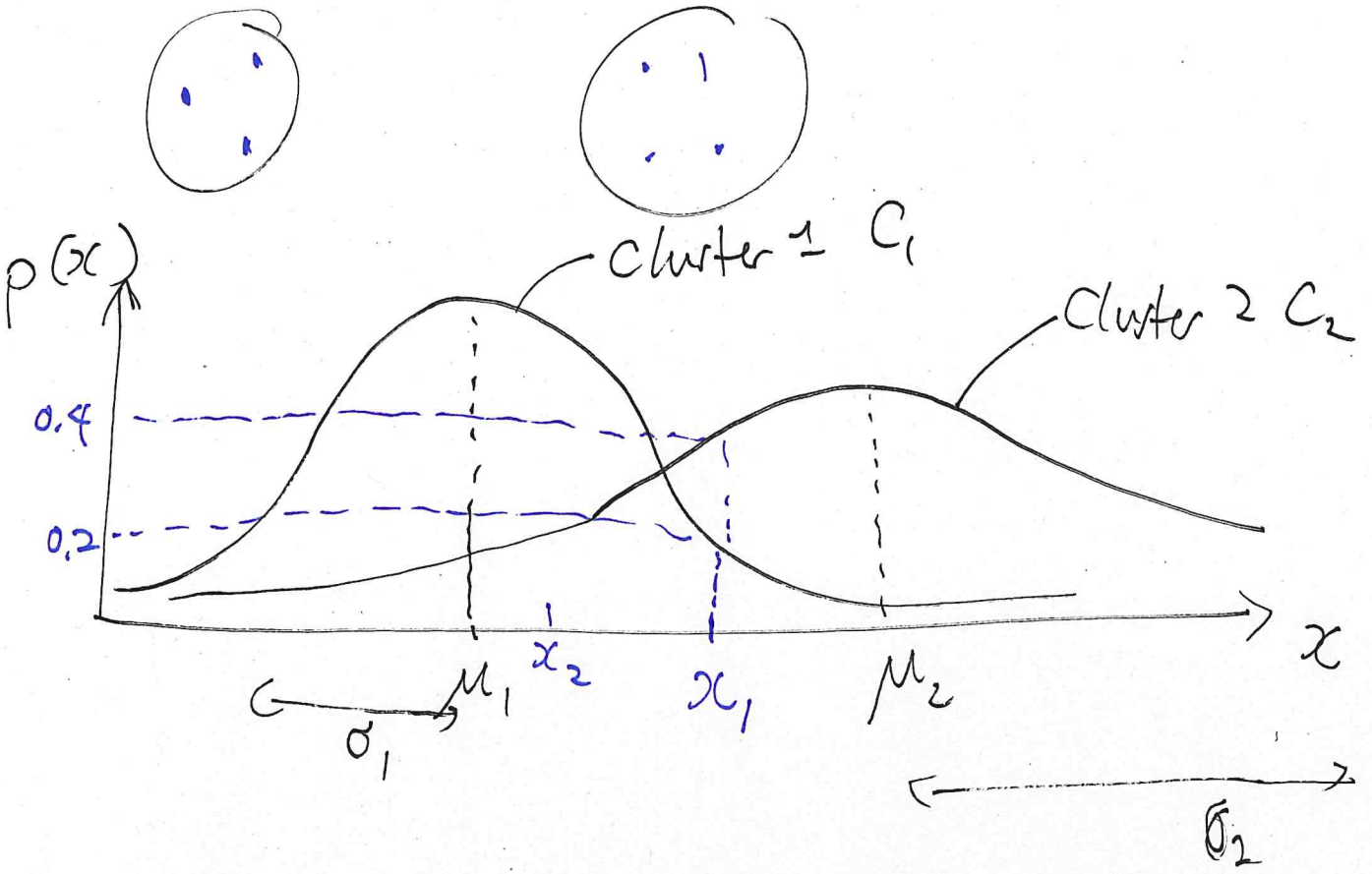
COMP5331 (25 Sept)

RM



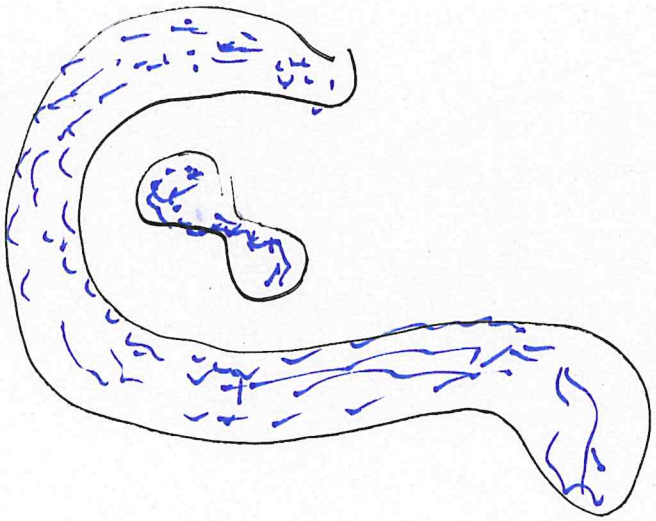
$$\rightarrow P(x_1 \in C_1) = \frac{0,2}{0,2+0,4} = \frac{1}{3}$$

$$P(x_1 \in C_2) = \frac{0,4}{0,2+0,4} = \frac{2}{3}$$

$$\rightarrow P(x_2 \in C_1) = \frac{1}{3}$$

$$P(x_2 \in C_2) = \frac{2}{3}$$

$$M_1 = \frac{P(x_1 \in C_1)}{P(x_1 \in C_1) + P(x_2 \in C_1)} \cdot x_1 + \frac{P(x_2 \in C_1)}{P(x_1 \in C_1) + P(x_2 \in C_1)} \cdot x_2$$



MinPts = 4

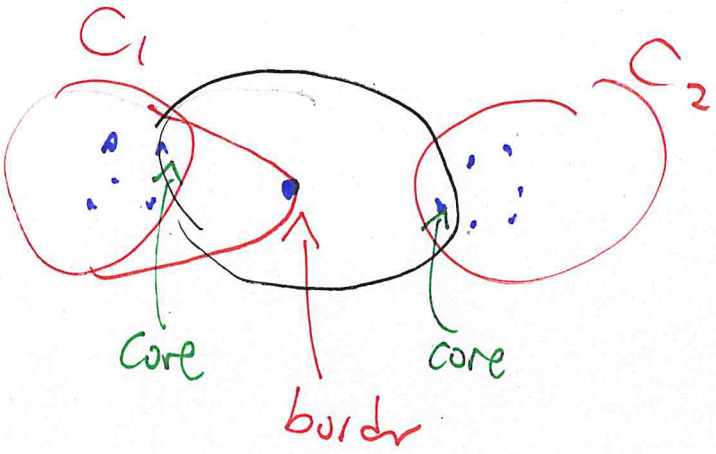
$N(a) = \{a\}$ ← noise

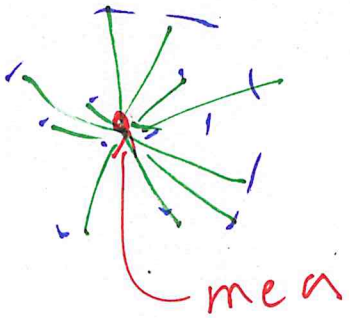
$N(b) = \{b, c, e\}$ ← border

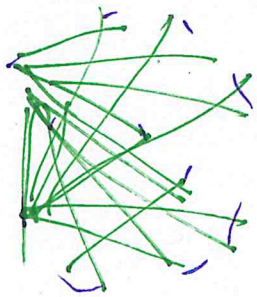
$N(c) = \{b, c, d, e\}$ ← core

$N(d) = \{c, d, e\}$ ← border

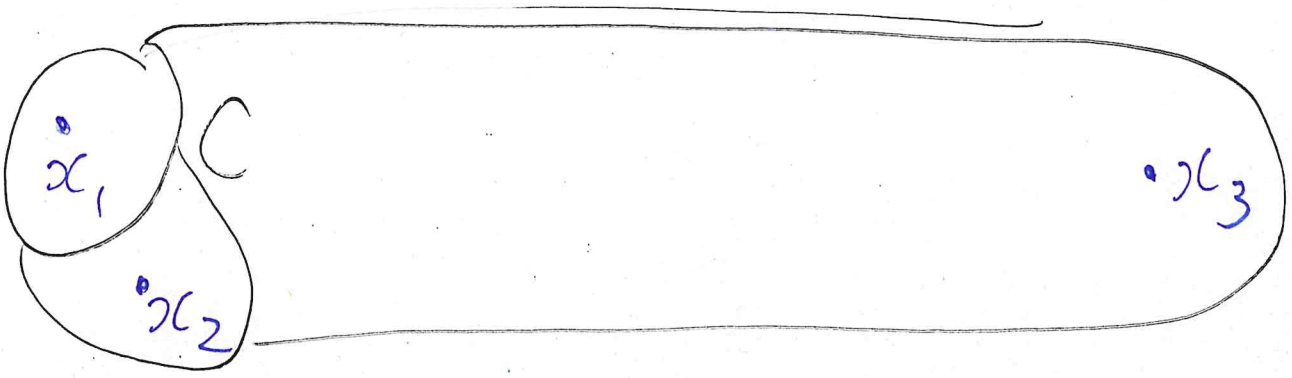
$N(e) = \{b, c, d, e\}$ ← core

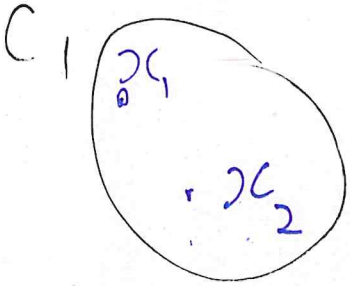


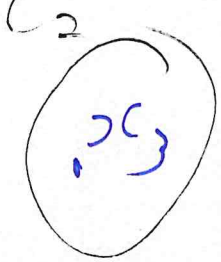




$L = (C_1, C_2)$







$$LS = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$SS = \begin{pmatrix} 1 \\ 2 \end{pmatrix}^2 + \begin{pmatrix} 3 \\ 4 \end{pmatrix}^2$$

$$= \begin{pmatrix} 1^2 \\ 2^2 \end{pmatrix} + \begin{pmatrix} 3^2 \\ 4^2 \end{pmatrix}$$

$$= \begin{pmatrix} 1^2 + 3^2 \\ 2^2 + 4^2 \end{pmatrix}$$

Express R and D in terms of n, LS, and SS

$$R = \sqrt{\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n}}$$

$$= \sqrt{\frac{\sum_{i=1}^{n}(x_i^2 - 2x_i\mu + \mu^2)}{n}}$$

$$= \sqrt{\frac{\sum_{i=1}^{n} x_i^2 - 2\mu\sum_{i=1}^{n} x_i + \sum_{i=1}^{n}\mu^2}{n}}$$

$$= \sqrt{\frac{SS - 2\mu \cdot LS + n\mu^2}{n}}$$

(by ① and ② and ③)

$$= \sqrt{\frac{SS - 2\cdot\frac{LS}{n}\cdot LS + n\left(\frac{LS}{n}\right)^2}{n}}$$

(by ④)

$$= \sqrt{\frac{SS - 2\cdot\frac{LS^2}{n} + \frac{LS^2}{n}}{n}}$$

$$= \sqrt{\frac{SS - \frac{LS^2}{n}}{n}}$$

$$= \sqrt{\frac{SS}{n} - \frac{LS^2}{n^2}}$$

$O(1)$

$$D = \sqrt{\frac{\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2}{n(n-1)}}$$

$$= \sqrt{\frac{\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i^2 - 2x_i x_j + x_j^2)}{n(n-1)}}$$

$$= \sqrt{\frac{\sum_{i=1}^{n}\left[\sum_{j=1}^{n} x_i^2 - 2x_i\sum_{j=1}^{n} x_j + \sum_{j=1}^{n} x_j^2\right]}{n(n-1)}}$$

$$= \sqrt{\frac{\sum_{i=1}^{n}\left[n \cdot x_i^2 - 2x_i \cdot LS + SS\right]}{n(n-1)}}$$

(by ① and ② and ③)

$$= \sqrt{\frac{n\sum_{i=1}^{n} x_i^2 - 2 \cdot LS \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} SS}{n(n-1)}}$$

$$= \sqrt{\frac{n \cdot SS - 2 \cdot LS \cdot LS + n \cdot SS}{n(n-1)}}$$

(by ① ② and ③)

$$= \sqrt{\frac{2n \cdot SS - 2 \cdot LS^2}{n(n-1)}}$$

$O(1)$

① $SS = \sum_{i=1}^{n} x_i^2$

② $LS = \sum_{i=1}^{n} x_i$

③ $\sum_{i=1}^{n} 4 = 4n$

4 ↑ constant　　4 ↑ constant

$\underbrace{4+4+\cdots+4}_{n \text{ times}}$

④ $\mu = \frac{LS}{n}$